NOVA IMS Information Management School

MDSAA

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Data Science and Advanced Analytics

PREVIEW

Lakehouse Data Architecture

Data as a first-class citizen within an organization

Fábio Rafael Santos Lopes

Internship Report

presented as partial requirement for obtaining a Master's Degree in Data Science and Advanced Analytics

NOVA Information Management School
Instituto Superior de Estatística e Gestão de Informação

Universidade Nova de Lisboa

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by

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Internship Report presented as partial requirement for obtaining the Master's degree in Data Science and Advanced Analytics, with a specialization in Data Science

Supervised by

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November, 2023

STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Lisbon, 21th of November of 2023

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DEDICATION

For Ana, whose support made these words possible.

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I stand at the culmination of a significant chapter in my academic journey, and it is with immense gratitude that I acknowledge those who have been instrumental in guiding me to this point.

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ABSTRACT

This thesis presents an in-depth exploration of the Lakehouse Data Architecture. This paradigm merges the strengths of data lakes and data warehouses, enabling organizations to harness the full potential of their data. The research investigates the architectural components, operational mechanisms, and strategic implications of implementing a Lakehouse within an organization using advanced technologies like Microsoft Azure, Google Cloud Platform, Databricks, Apache Spark, Delta Lake, and Dremio. The study also scrutinizes the Lakehouse's ability to facilitate a data-centric culture by integrating advanced analytics into business processes. The thesis further delves into the FAIR data principles, advocating for data to be Findable, Accessible, Interoperable, and Reusable, and the Data Mesh concept, a decentralized data management approach. The research concludes that the Lakehouse architecture provides a comprehensive and robust framework for managing vast and diverse data sets, optimizing data pipeline performance, reducing redundancy, and enhancing data security. It underscores the pivotal role of the Lakehouse in driving strategic innovation and positions it as a flexible and adaptable model for future technological advancements in AI and machine learning. The insights offered in this thesis serve as a guide for organizations aiming to navigate the complexities of becoming data-centric and underscore the transformative power of modern data platforms.

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KEYWORDS

Data Mesh; Databricks; Data Lakehouse; Delta Lake; Microsoft Azure; Apache Spark

Sustainable Development Goals (SDG):

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LIST OF ABBREVIATIONS AND ACRONYMS

GCP	Google Cloud Platform
AWS	Amazon Web Services
EMR	Elastic Map Reduce
GCS	Google Cloud Storage
DBU	Databricks Unit
EAN	European Article Number
UPC	Universal Product Code
FPI	Financial Performance Indicator
AAD	Azure Active Directory
RBAC	Role-Based Access Control
CDC	Change Data Capture
API	Application Programmable Interface
ISO	International Organization for Standardization
MLOPS	Machine Learning Operations
IaaS	Infraestruture as a Service
PaaS	Platform as a Service
HDFS	Hadoop Distributed File System
AI	Artificial Intelligence
ML	Machine Learning

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1. INTRODUCTION

In the digital era, data is often referred to as the new oil (The Economist, 2017), an invaluable asset that can drive innovation, strategic decision-making, and competitive advantage (Chen et al., 2012a). However, the potential of data can only be fully harnessed when treated as a first-class citizen within the organizational ecosystem (Alaimo & Kallinikos, 2022). This report explores the Lakehouse Data Architecture, a new paradigm that promises to elevate data to the forefront of an organization, ensuring its accessibility, usability, and integrity across diverse business functions (Zaharia et al., 2021).

The impetus for the Lakehouse model emerges from the convergence of two traditionally distinct data management systems: the data warehouse, known for its structured, curated datasets ideal for analytical workloads (Chaudhuri & Dayal, 1997), and the data lake, a more flexible repository capable of storing vast amounts of raw, unstructured data (Miloslavskaya & Tolstoy, 2016). The Lakehouse architecture embodies the strengths of both, providing a unified platform that supports a wide array of data types and workloads, from machine learning to business intelligence (Zaharia et al., 2021).

This study delves into the architectural components, operational mechanisms, and strategic implications of implementing a Lakehouse within an organization. It scrutinizes the architecture's ability to facilitate a data-centric culture by integrating advanced analytics into the business processes. The thesis evaluates the role of cutting-edge technologies in realizing the Lakehouse vision, including Microsoft Azure, Google Cloud Platform, Databricks, Apache Spark, Delta Lake, Dremio, and the emerging concept of a Data Mesh.

In a landscape where data architectures are rapidly evolving (Serra, 2024), this thesis stands as a testament to the innovative spirit of the field. It provides a comprehensive technical review of the Lakehouse architecture and situates it within a practical business context. The insights offered are targeted to guide organizations toward becoming truly data-driven, underscoring the transformational power of modern data platforms.

As we navigate through the chapters, we will uncover the foundational framework of the Lakehouse, discover the journey of the organization in study through different technologies, the challenges of the current architecture, the realization of a Lakehouse architecture, and two successful use cases put in place.

As a Solutions Architect, I helped define the organization's data strategy, drawing on the comprehensive knowledge from my Master's program. Alongside the team of Solution Architects, I orchestrated the design of our Lakehouse Data Architecture, engaging closely with business stakeholders to ensure their needs were met and with engineering teams to guide the development of the data platform. This synergy was critical for a scalable system prepared for future data management challenges. My academic expertise, especially in Big Data Analytics, Big Data Modelling and Management, and Storing and Retrieving Data, was crucial in successfully implementing a robust data architecture that supports diverse analytical workloads and the company's strategic objectives.

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2. FRAMEWORK & TECHNICAL REVIEW

2.1. TECHNICAL REVIEW

2.1.1. Lakehouse Architecture

The world of data management has witnessed significant shifts over the years, with data lakes and data warehouses being primary staples for organizations (Arif & Mujtaba, 2015; Khine Pwint Phyu & Wang Zhao Shun, 2018).

Organizations have used data warehouses since the early 1990s and are a cornerstone of the world of data management (Jarke et al., 2002). A data warehouse is a specialized system that supports business intelligence activities, primarily analytical processing, reporting, and data retrieval (Chaudhuri & Dayal, 1997). Unlike data lakes, which store raw data, data warehouses contain structured, processed, and aggregated data.

The primary advantage of data warehouses lies in their optimized performance since they are built to handle complex queries swiftly, ensuring that businesses can derive insights and make decisions on time (Garani et al., 2019). Their structured nature ensures data consistency, reliability, and accuracy, making them a trusted source for critical business operations.

However, this structured approach brings its own set of challenges. Data warehouses require a significant upfront investment in design and architecture (Subramanian et al., 1997). The need for data to fit into predefined schemas can sometimes make them less adaptable to changing business needs or new data sources (March & Hevner, 2007). Scalability can be a concern, especially when dealing with vast and rapidly growing data volumes.

A data lake is a centralized repository that stores vast amounts of raw data, regardless of source or format. This can range from structured data from databases to unstructured data like images or videos.

Central to the data lake concept is the Hadoop Distributed File System, HDFS, a foundational storage system designed by Google to house vast amounts of data across distributed clusters of machines. Born from a paper that introduced the world to the Hadoop framework, HDFS was designed to be highly fault-tolerant, scalable, and suitable for applications with large data sets (Ghemawat et al., 2003). It breaks down large files into blocks, distributing them across nodes in a cluster, ensuring data redundancy and resilience.

The primary benefit of data lakes is their inherent flexibility. Organizations can store data in the lake without structure or processing, making it a scalable solution for ever-growing data volumes (Miloslavskaya & Tolstoy, 2016). This flexibility also allows for diverse analytical tools and methods to be applied, catering to varied business needs.

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